River-avulsion fingerprints from published image stacks and channel masks

Independent computational reproduction of Valenza et al. (2020)

October 2026

Abstract

River avulsions disturb different amounts of floodplain while establishing a new flow path. Valenza et al. (2020) quantify this disturbance using annual satellite observations and manually delineated channel masks. Here we test the downstream calculation using the published processed images and masks, with an independent C++ implementation and original plotting scripts. This experiment accepts the authors' preprocessing and delineation as input; it tests the subsequent spectral filtering, temporal accumulation and area ratio calculation. Published measurements of channel geometry and morphology are excluded from the analysis inputs. We calculate defined ratios for 63 events, spanning 0.03879–463.9, compared with the published range of 0.4–42.4. Thus the primary calculation does not reproduce the reported range. Explicit parent-window alternatives materially change the result. Independent geometry covers only four illustrative examples, so the population relationships remain untested. The panel ledger and execution record identify the limits of this result. A successful software build does not by itself establish reproduction of the scientific results.

1 Question and scope

The observable is the area of floodplain activity associated with channel relocation. Its normalization by the area of the former channel defines a dimensionless avulsion style ratio. A large ratio indicates a large disturbed area relative to that channel; it does not by itself establish the mechanism responsible for the disturbance.

The previous exercise began with independently acquired Landsat observations. This exercise begins with the processed image stacks and parent/avulsion masks deposited by Valenza and colleagues. The scientific specifications remain the paper and its written supplementary material. No author MATLAB, Earth Engine, analysis or plotting script is opened or executed. No published plot is digitized, and no published figure is substituted for a generated result.

The user also permitted interpretation metadata: site identities and coordinates, band/year mappings, selected bands and event-onset indices. Published widths, slopes, distances, morphology classes, fingerprint areas, style ratios and signature choices remain excluded from analysis inputs. The metadata extractor reads only the authorized spreadsheet columns and retains their source-row numbers. It does not retain the source workbook.

Independence is limited to the downstream implementation. The accepted masks contain the authors' observational judgments about event extent and unrelated water bodies. The processed images contain their cloud screening, annual compositing and tasseled-cap transformation. Those operations are not newly validated by matching a downstream fingerprint. Published numerical values were seen during the earlier feasibility discussion and while extracting the supplement as text. They were not copied into analysis inputs or used to tune thresholds, windows or masks. This is not a blinded experiment.

2 Data and mathematical specification

The institutional archive supplies annual brightness, greenness and wetness GeoTIFF stacks, together with region-of-interest masks. The archive period is 1986–2017. The paper describes Landsat observations with nominal 30 m sampling. We preserve each supplied grid rather than silently imposing a different one. The figure captions distinguish these accepted observations from quantities calculated by the new implementation.

2.1 Annual classification

Let $B_y(x)$, $G_y(x)$ and $W_y(x)$ be the three transformed components at pixel x in annual layer y. We retain pixels with valid finite values in all three components. File-defined nodata and validity masks are respected; a finite zero is not automatically treated as missing. Across the retained pixels in each complete image domain, calculate

$$I_{B,y}(x) = \mathbf{1}\{B_y(x) > Q_{0.90}(B_y)\}\mathbf{1}\{G_y(x) < Q_{0.50}(G_y)\}, \tag{1}$$

$$I_{W,y}(x) = \mathbf{1}\{W_y(x) > Q_{0.75}(W_y)\}\mathbf{1}\{G_y(x) < Q_{0.50}(G_y)\}. \tag{2}$$

Here Q_p is the sample percentile. These rules identify unusually bright or wet surfaces with low greenness. Both inequalities are strict. A constant scene therefore contributes no detections.

We use linear interpolation at index $(n-1)p$ in a zero-indexed sorted sample of n valid values, retaining the convention used in the first exercise. The written description does not specify a percentile interpolation rule. The thresholds are recalculated from the supplied annual images and are saved with valid-pixel and detection counts. They are not fitted to published areas or ratios.

2.2 Temporal accumulation and area

For a signature s and a channel region r, calculate

$$F_{r,s}(x) = M_{r,s}(x) \sum_{y \in \mathcal{Y}_r} I_{s,y}(x), \qquad N_{r,s} = \sum_{x} \mathbf{1}\{F_{r,s}(x) > 0\}. \tag{3}$$

The accepted binary mask M restricts activity to the indicated channel region. A pixel detected in several years contributes once to the area count, while its cumulative raster value records the number of detections. We also save the number of valid observation years at each masked pixel. Thus missing observations remain distinguishable from observed absence.

For a complete event, the style ratio is

$$S_{\mathrm{R},s} = \frac{N_{\mathrm{avulsion},s}}{N_{\mathrm{parent},s}}. \tag{4}$$

A zero parent count gives an undefined ratio. Where both signatures have adequate input coverage, the selected signature minimizes the avulsion fingerprint area. This differs from minimizing the ratio when the parent counts differ. Exact ties select brightness, consistent with the first exercise. Both signature-specific counts remain available for inspection.

Pixel counts follow the published definition. The determinant of the image transform is recorded as a native-coordinate pixel area. An angular pixel area is not interpreted as square metres. Multiple image parts require explicit accounting before their counts can be combined; overlap and missing parts must not silently inflate or truncate an event.

2.3 Time-window interpretation

The authorized metadata provide a list of selected native bands and the position of the onset of avulsion activity. The onset index refers to the selected list, as shown by cases whose onset index does not occur in the native-band list. With layers ordered from newest to oldest, positions up to and including the onset index form the avulsion window; older positions form the parent window. This interpretation is inferred from the written metadata, without consulting the authors' program.

Some year labels disagree with the selected bands. These inconsistencies are preserved in an audit table. Native band indices define the numerical calculation; unresolved calendar labels are not silently replaced. Multiple onset indices require separate timing scenarios. The exact band assignments and any additional decisions are saved with the configuration.

Image part	Metadata issue
V14	Year count differs from band count; intended year list unresolved.
B1_b	Year count differs from band count; intended year list unresolved.
B14_b	Multiple onset indices; evaluate both as separate timing scenarios.
H2	Year count differs from band count; intended year list unresolved.
G4	Invalid year range; preserve raw text and use supplied band indices. Year count differs from band count; intended year list unresolved.
G15	Year label outside 1986–2017; label not used to change band selection.

Native image labels provide calendar names without changing the selected bands. They do not resolve which conflicting list the authors intended.

2.4 Geometry and statistical relationships

The paper's other observables are average channel width near the avulsion site, along-channel distance to a physiographically identified mountain front, parent-channel slope along a sufficiently long reach, and flow-thread counts at multiple cross sections. The reported slope measurements usually span 5–10 km and sometimes longer. A mask surrounding a channel is not a measurement of these quantities.

The statistical implementation accepts independently documented geometry, calculates normalized distance D/w, and fits a line in logarithmic space using only $D/w > 60$. Log-space ordinary least squares is an explicit assumption because the paper does not fully specify its regression convention. Group summaries use linearly interpolated quartiles and range whiskers. Missing measurements remain absent. A spectral footprint alone does not determine a channel's morphology class.

The four illustrative geometry observations are saved as explicit pixel-space annotations. Width sampling integrates the wet-pixel intersections of three short cross sections per event; a 1 m integration interval does not improve the 30 m image resolution. A section must have valid observations and leave the wet region at both ends. We assign a coarse morphology class only from the independently recorded visible flow-thread counts. These observations are single-date examples and may miss narrow threads.

We traced three example parent-channel or channel-belt centerlines before calculating style ratios, sampled independently acquired Terrain Tiles elevation at 30 m intervals, and estimated downstream slope by ordinary least squares. We also save endpoint slopes. The illustrative logarithmic comparison retains only profiles at least 5 km long with complete terrain support and positive slope under both estimates. All attempted estimates, including rejected ones, remain in the geometry table. This explicit screen is a limited consistency check, not an elevation-error model. Vegetation, DEM source differences, centerline placement and the short sample prevent equating these estimates to the paper's slope measurements.

3 Calculated results and panel assessment

We calculated 77 image-part/timing jobs for 63 events. 63 events have defined style ratios under the stated primary assumptions. The calculated range is 0.03879–463.9, from G23 to B10. Signature selection used the newly calculated avulsion areas. It selected brightness for 57 events and wetness for 6 events.

Every deposited event mask is named for only one spectral signature. We therefore apply both filters to the same supplied geometry, retaining the mask's source name only as provenance. This tests numerical signature selection on fixed geometry; it cannot recreate a separately redrawn mask for the alternative signature.

The embedded image calendar labels disagree with 33 selected metadata labels across 6 jobs. Native band selections remain unchanged. The file `temporal_labels.csv` records every assignment. The unused-input ledger includes B5, which lacks a corresponding selected-window record, and extra B10 tiles. No excluded measured result is used to fill these gaps.

The overlap audit identifies 15 site/signature/role combinations with duplicate detected pixels. The primary event counts take the spatial union on aligned native grids, so a physical pixel is counted once. A separate summed-part calculation quantifies the effect of simply adding tile counts. Mosaic colors use the maximum per-part detection count in overlaps; heterogeneous windows prevent interpreting that color as one uniform observation period.

Across the 146 primary image-part/role windows, the fraction of mask pixels observed at least once ranges from 91.59 to 100 percent; 22 windows have some never-observed pixels. This cumulative coverage does not imply full coverage in every year. Missing pixels are not filled, and the ratio is not corrected by an assumed missing-area multiplier.

We independently annotated 4 prespecified illustrative events (B11, B12, V11 and V15). These are single-date, 30 m observations, not a new 63-event morphology survey. The elevation exercise produced 2 positive slopes meeting the explicit reach-length and sign checks. Widths are wet-pixel cross-section estimates with a native pixel scale of 30 m. No mountain-front distance was measured. Consequently, the full distance relationship and the published population distributions remain unreproduced.

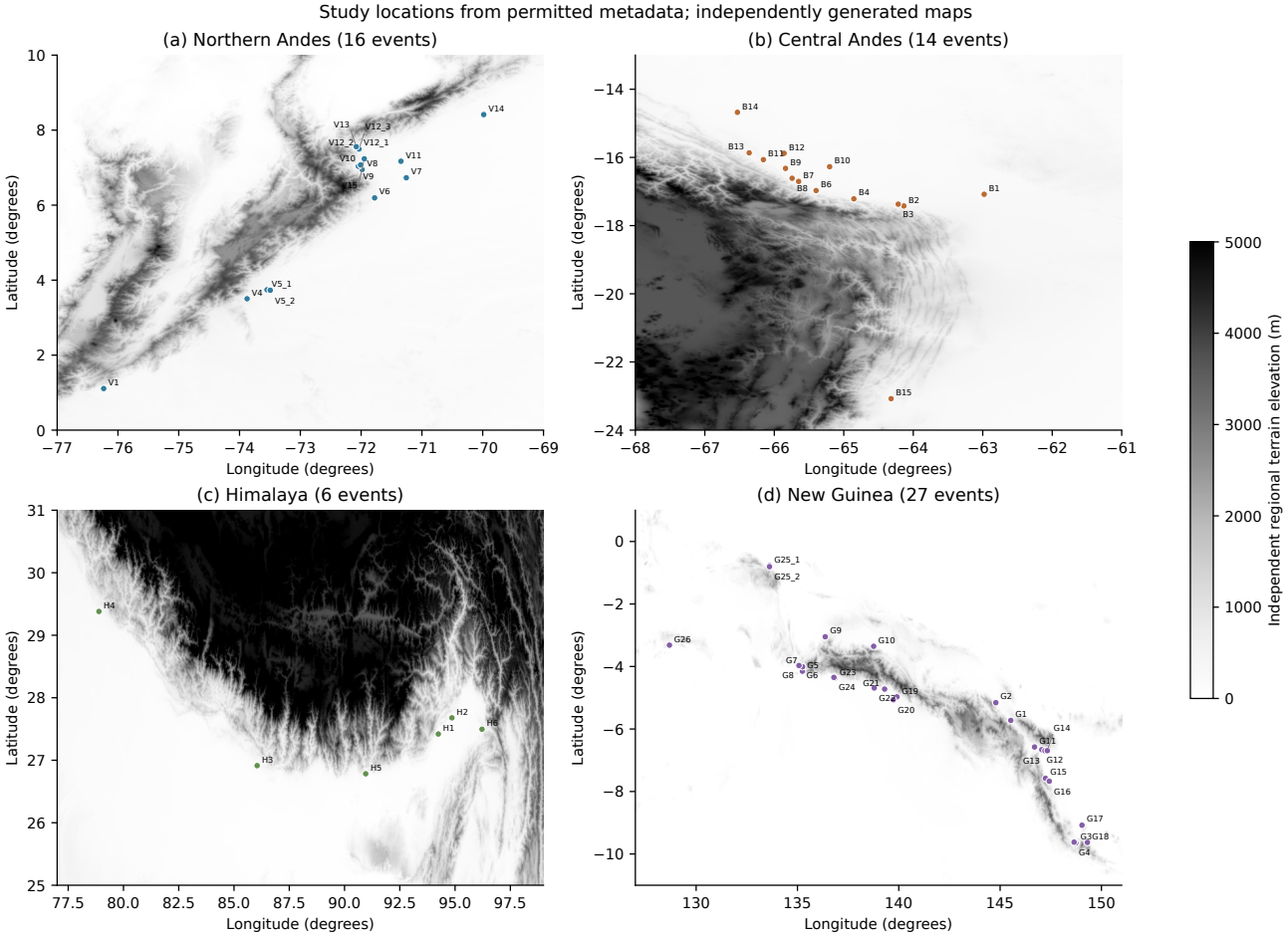


Figure 1: Reconstructed geographic site distributions for (a) the northern Andes, (b) the central Andes, (c) the Himalaya and (d) New Guinea. Coordinates are permitted source metadata; the terrain background comes from the independent public terrain acquisition used in the first exercise. B15 is included in this map. The map reproduces geographic content rather than the publication's exact cartographic design.

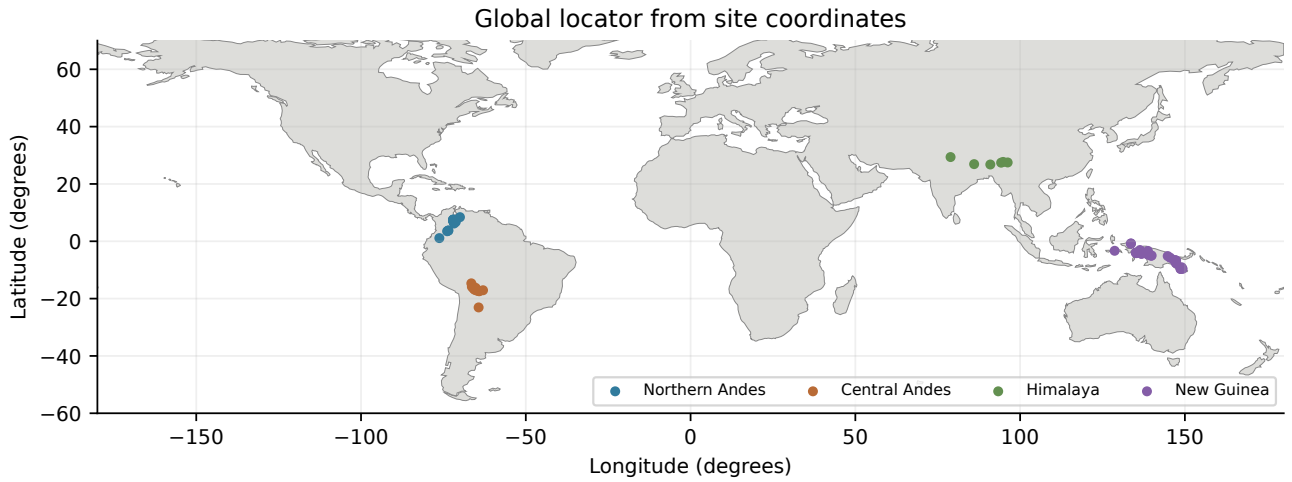


Figure 2: Global study-site positions on geographic axes. The continental outline is independent Natural Earth vector data. This generated locator uses a simpler design than the published inset.

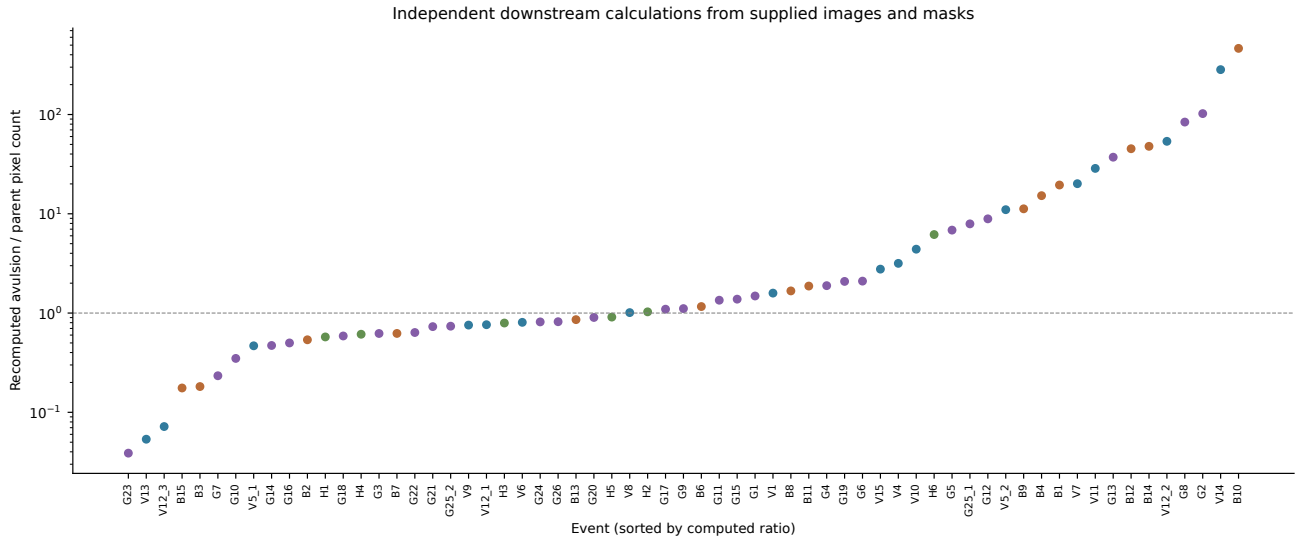


Figure 3: Style ratios calculated from the supplied image stacks and masks by the new implementation. Each point is an event with a defined ratio; sorting uses the computed ratio. Color denotes the geographic prefix. The dashed line denotes equal avulsion and parent counts. This additional diagnostic is not a substitute for the paper's geometry or morphology relationships.

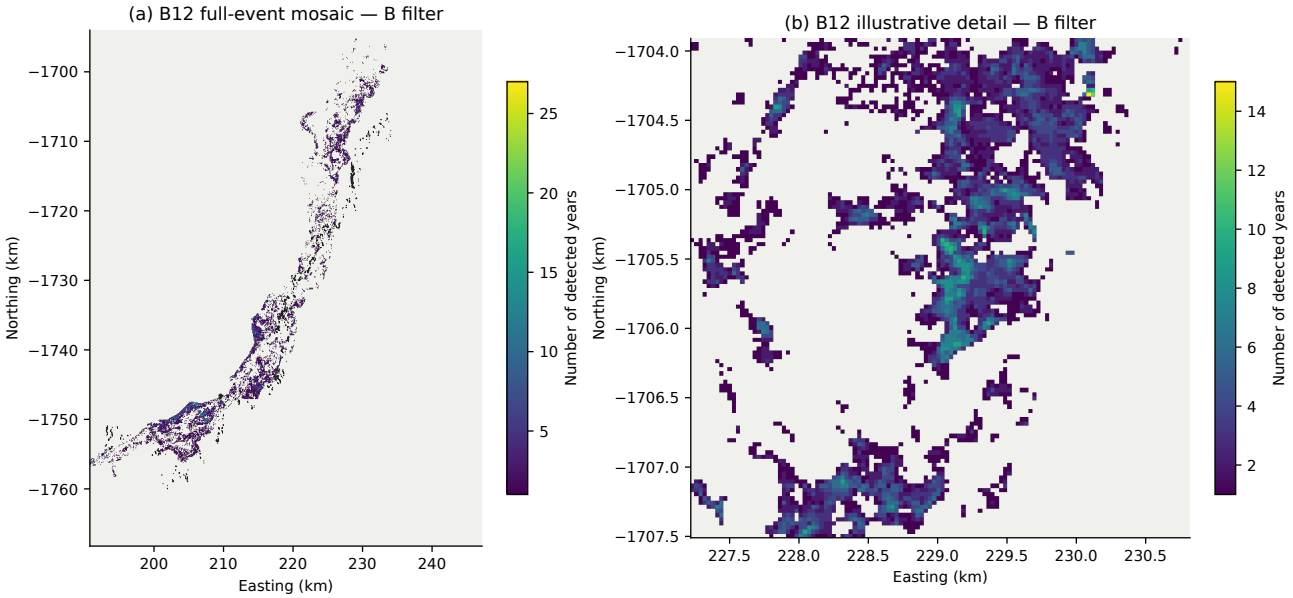


Figure 4: B12 full-event cumulative mosaic and illustrative detail from saved independent calculations. Color measures the number of detected years. Dashed contours, where present, outline the calculated parent fingerprint. The input images and region masks are supplied data. Exact spatial and temporal correspondence with the publication is not established: the full mosaic is larger than the published central-part view, and the selected detail is not the original validation window. Mosaic colors take the maximum per-part count in overlaps.

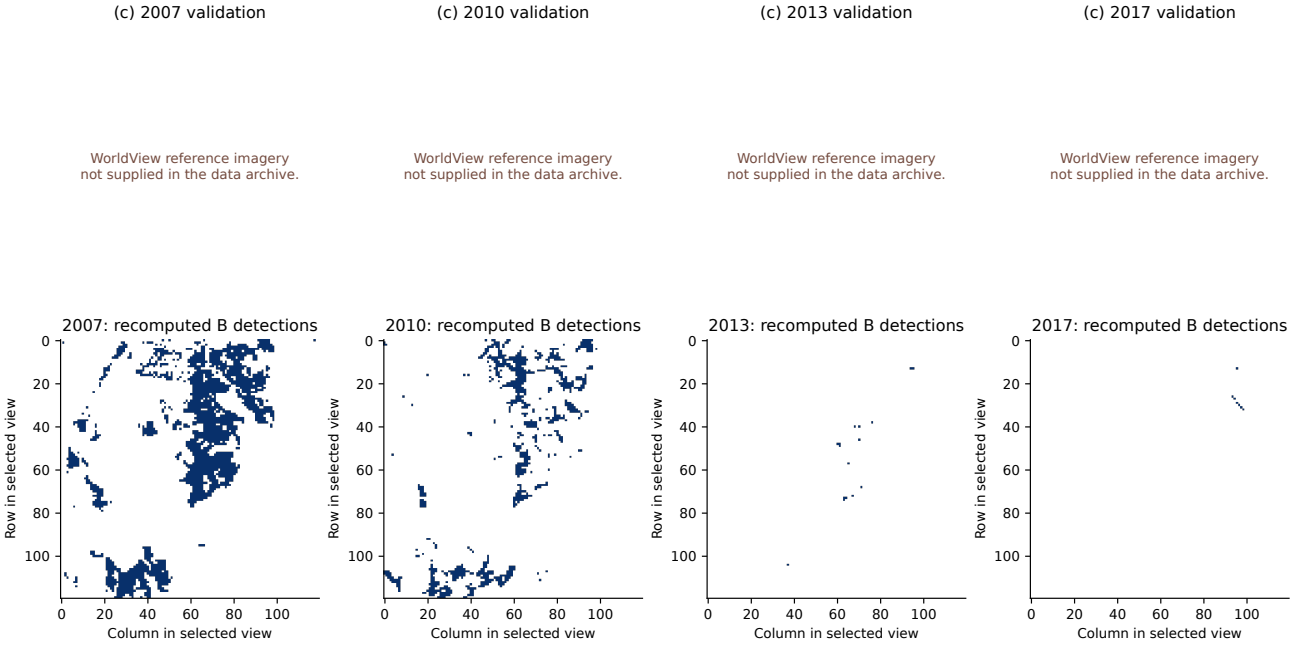


Figure 5: Attempted annual counterparts for Figure 2c. The lower row shows independently recomputed detections within the accepted avulsion mask. The upper row records the availability of the separate high-resolution validation observations. This independently chosen detail differs from the published validation window, so these panels do not reproduce that validation comparison. No published image is inserted as a replacement.

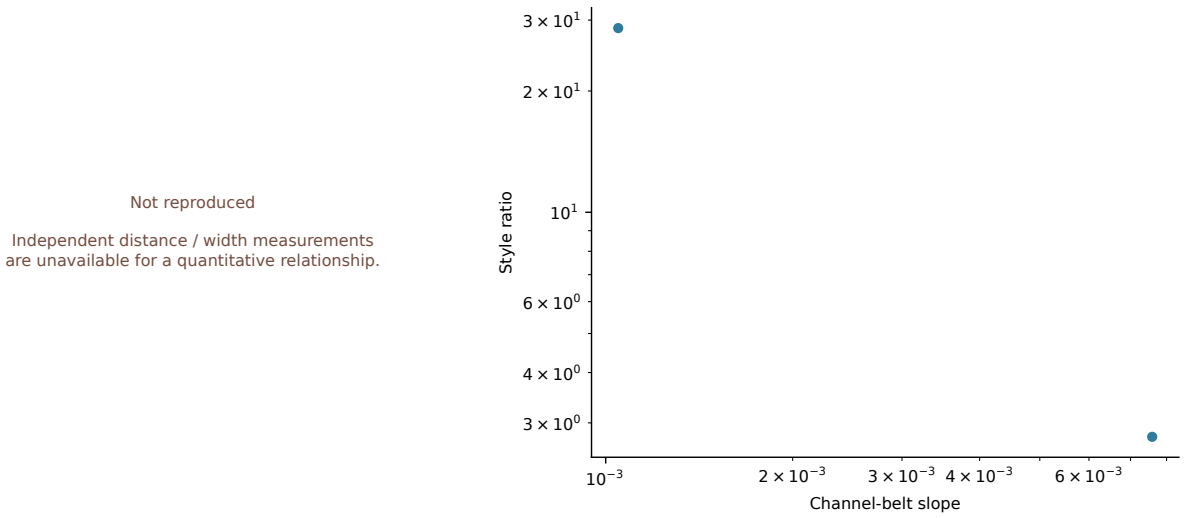


Figure 6: Attempt to calculate the published style-ratio relationships. Only independently documented geometry can define the horizontal coordinates. An absence notice denotes an unreproduced relationship, not a null result.

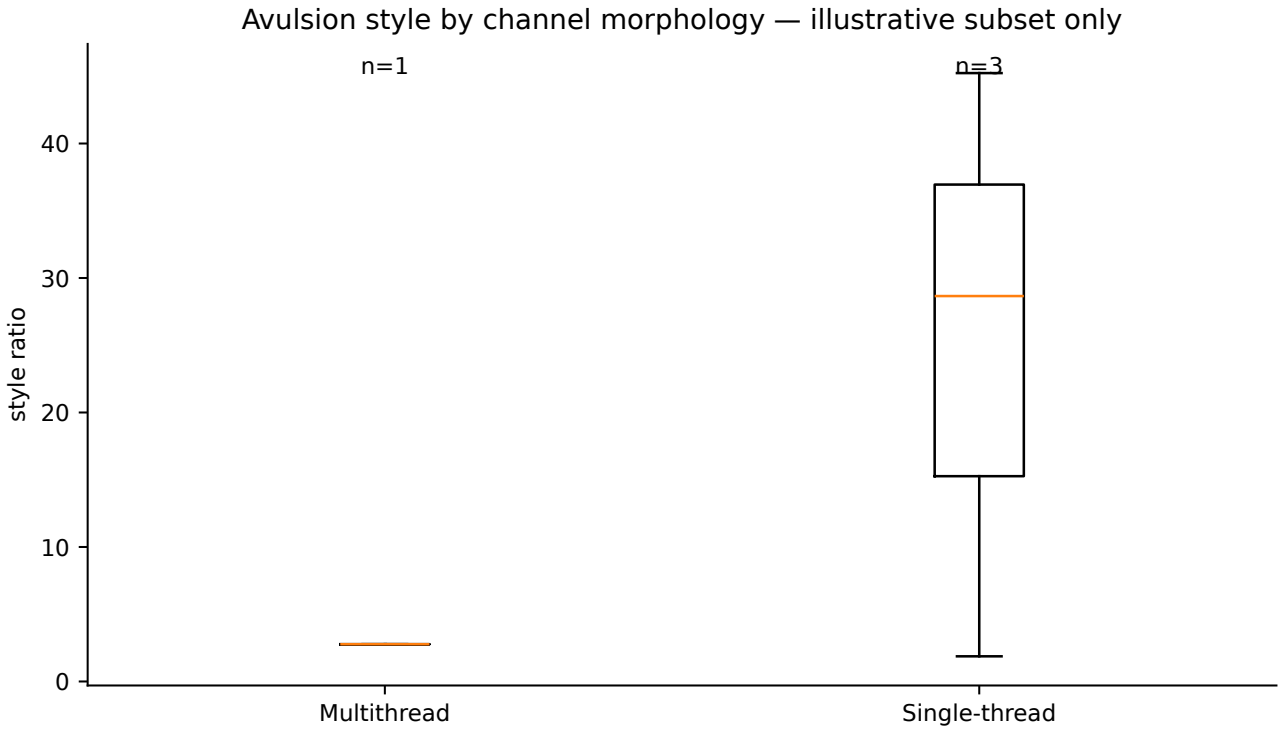


Figure 7: Attempt to reconstruct style-ratio distributions by independently measured morphology. Where data exist, boxes show quartiles and whiskers show the full range. Missing classifications are not inferred from the desired style ratio.

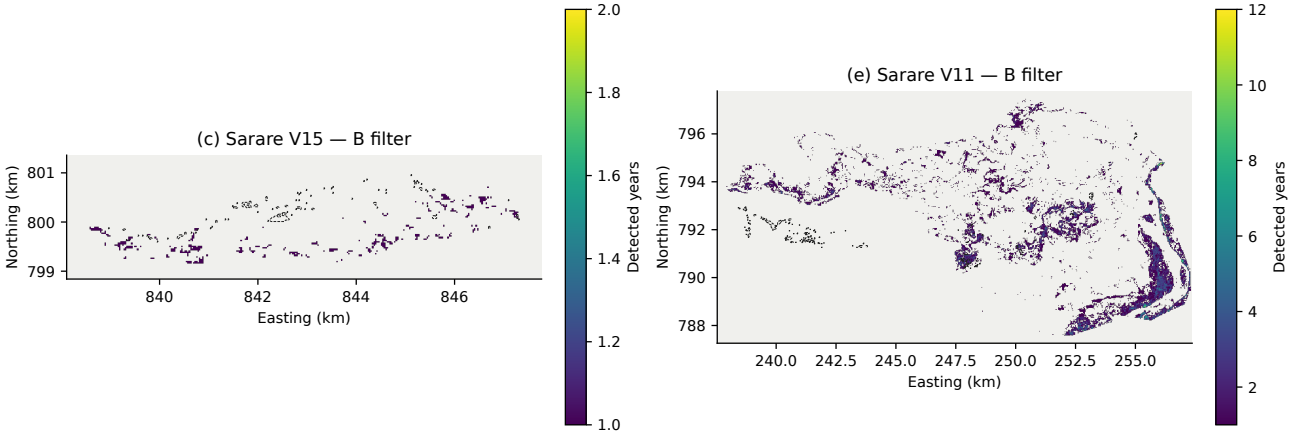


Figure 8: Independent fingerprint calculations for the two Sarare events used in Figure 5. Color denotes detected years; dashed parent contours are calculated from the corresponding observations. The accepted masks remove the need for new manual event delineation in this exercise.

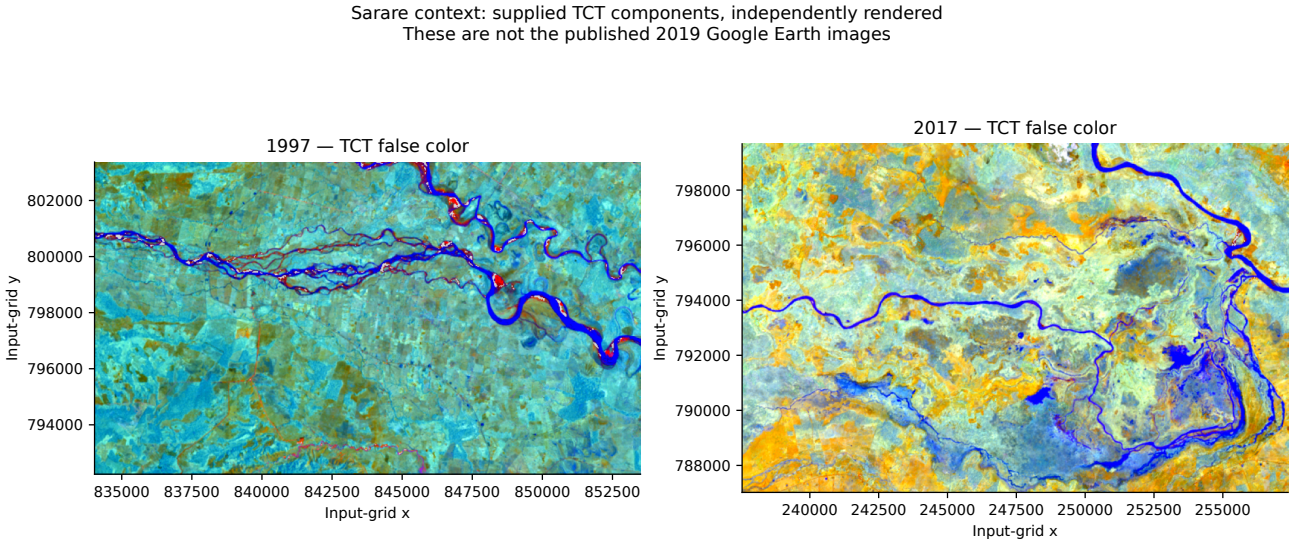


Figure 9: Generated context views for the Sarare events. Tasseled-cap components are displayed as false color where available and are explicitly distinguished from the paper's Google Earth imagery.

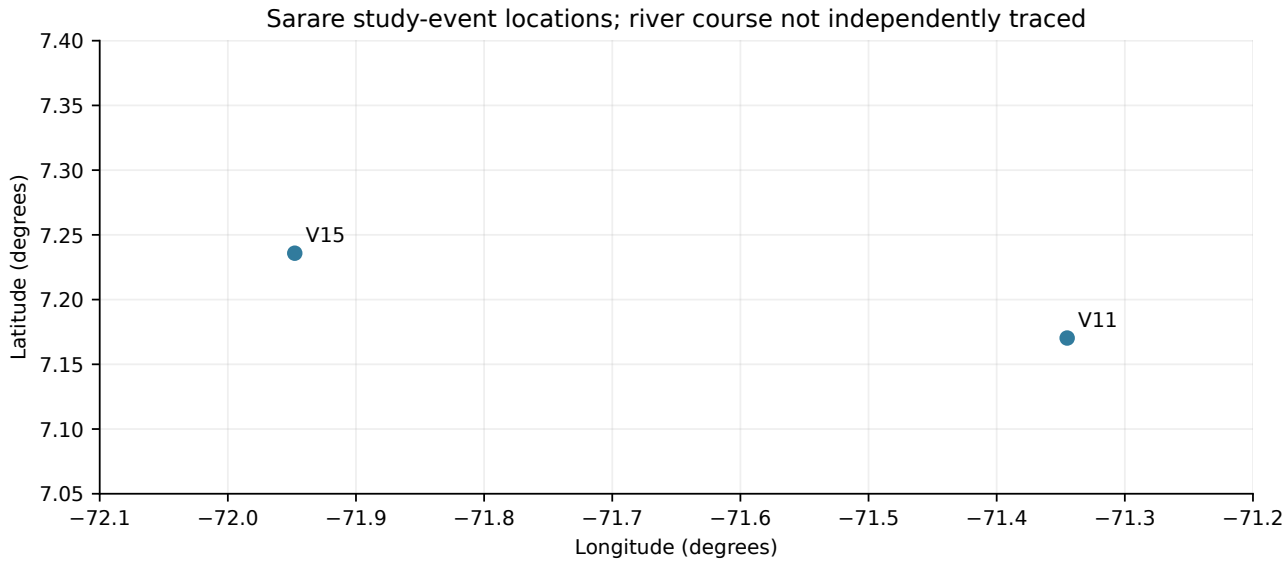


Figure 10: Independent geographic locator for the two Sarare events. The river course between the two deposited image domains has not been traced, limiting correspondence to Figure 5a.

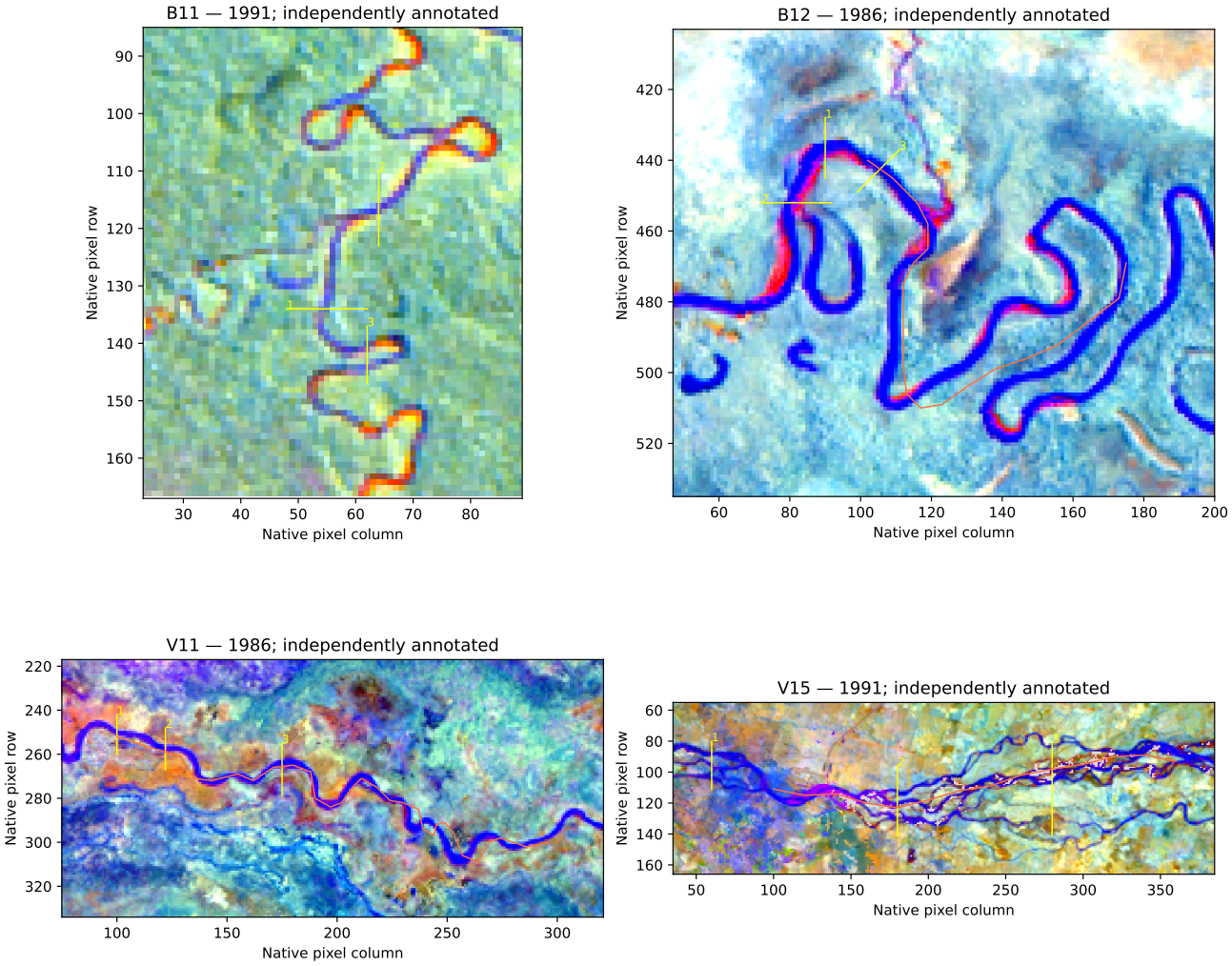


Figure 11: Four independent illustrative observations. Yellow lines mark the three width and flow-thread cross sections. Orange lines mark the traced parent-channel or channel-belt centerline where one was recorded. Backgrounds show the supplied TCT components with a 2–98 percent display stretch. Axes are native pixel coordinates; the annotations and all derived measurements are saved separately.

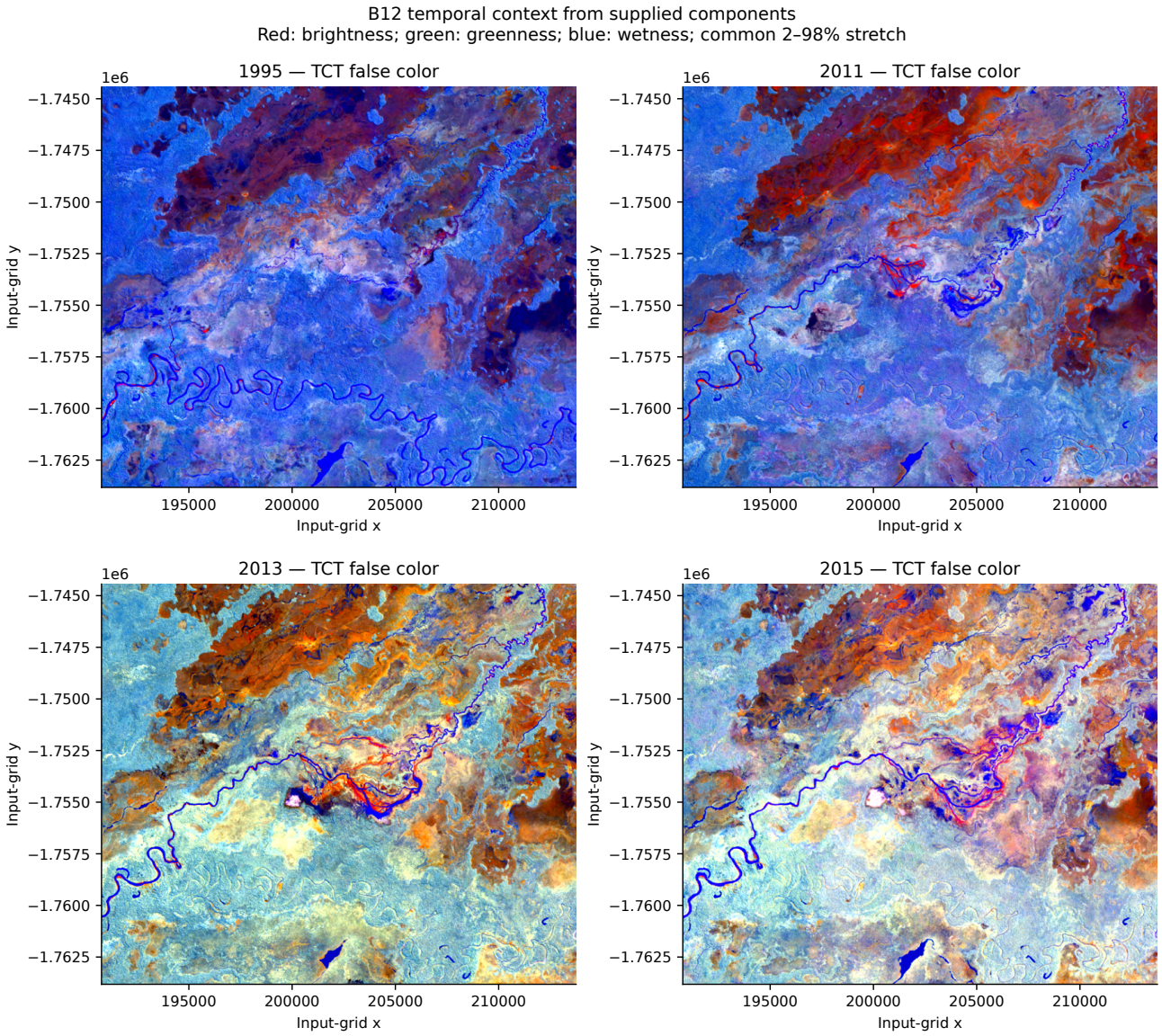


Figure 12: Temporal context for B12. These views use the supplied tasseled-cap components with a documented common display stretch. They are not reconstructed conventional RGB photographs.

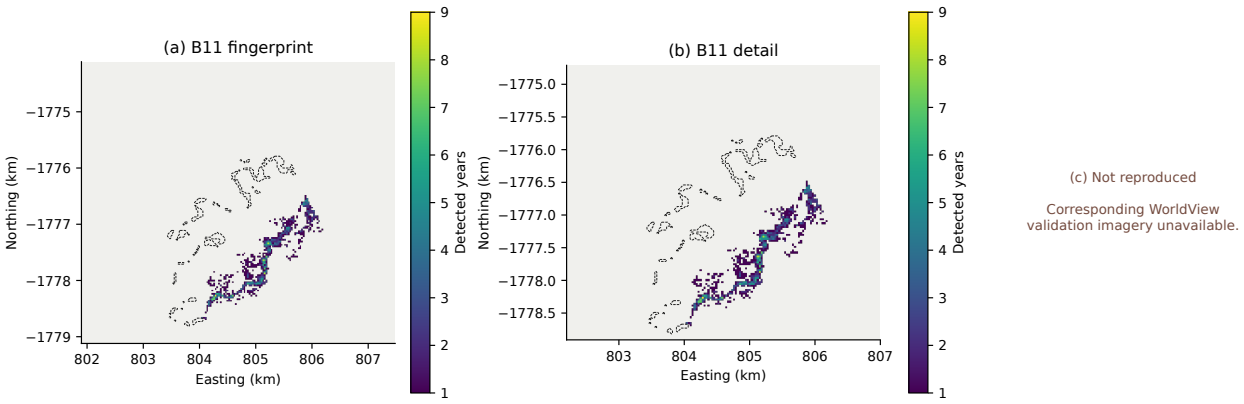


Figure 13: Independent B11 fingerprint and detail, with the separate high-resolution validation slot identified. Color records detected years.

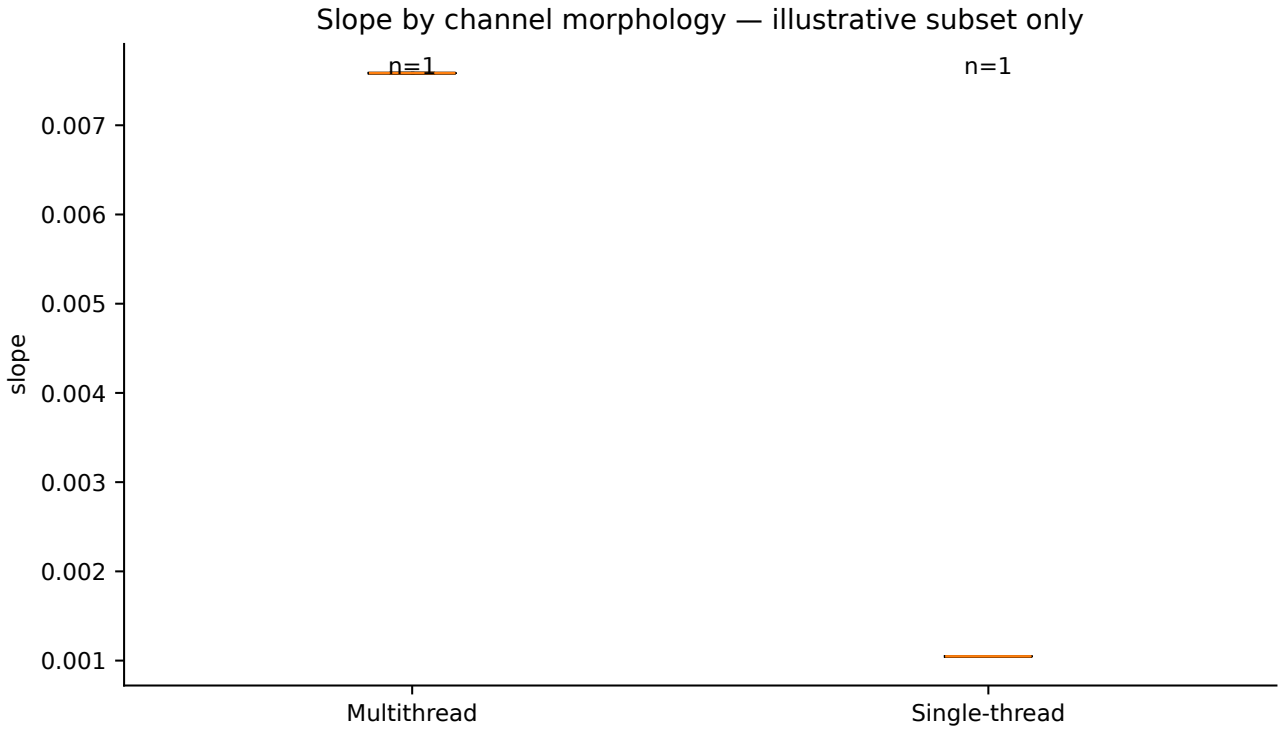


Figure 14: Attempted slope distributions by independently measured channel morphology. Missing observations are identified explicitly.

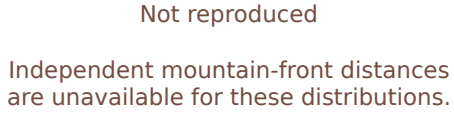


Figure 15: Attempted normalized-distance distributions by independently measured channel morphology.

4 Implementation, validation and reproducibility

The C++17 implementation uses GDAL for raster input and output. Numerical operations are implemented directly: percentile selection, strict spectral classification, temporal accumulation, union counts, ratio selection and statistical summaries. Python handles the authorized metadata, data-format conversion, original plotting and report assembly. No author's program is a build dependency.

Analytical checks cover constant scenes, missing values, singleton samples and known quantiles. Separate raster tests exercise nodata, binary masks, multiple years, temporal unions and rejection of shifted grids. Additional checks use known multi-part totals and exact power-law relationships to test signature selection and regression. These artificial fixtures are software validation cases, never study data or manuscript results.

Real-data checks independently sum saved annual detections and compare the result with the C++ cumulative rasters. They verify that detections are binary, fingerprints do not exceed valid observation years, and saved pixel counts agree with the rasters. These checks establish arithmetic consistency; they cannot independently validate the supplied masks or observational interpretation.

The recorded full-rebuild state is **completed**. The recorded calculation took 630.0 s. The implementation passed 378 analytical quantile checks, 105 separate raster-integration checks, 11 summary/regression checks and 8 exact mosaic checks. The real outputs passed 6974 arithmetic and raster-consistency checks. The command-by-command execution record is `data/results/rebuild_outcome.json`; logs and hashes are retained.

All project files, dependencies and caches are kept under a selected `/data/data[n]` directory. The institutional raw archive is retrieved through a documented download step and is not a substitute for saved generated outputs. Input hashes, file inventories, configuration and commands are retained. The README gives the exact complete rebuild procedure and distinguishes a computation using cached inputs from a new download.

5 Limits of the inference

Accepting processed imagery and channel masks removes several decisions that constrained the first exercise. Agreement after this input boundary would test downstream numerical operations, while leaving the correctness of the authors' preprocessing and manual masks outside the experiment. Disagreement may arise from documented ambiguities in percentile conventions, selected-band metadata, temporal splits, mask correspondence or multi-part accounting. These alternatives should be assessed through explicit sensitivity calculations, not by choosing the variant closest to a published ratio.

The core physical interpretation additionally depends on channel geometry and morphology. Recomputed fingerprints cannot establish those relationships without the corresponding independent observations. The panel ledger therefore separates calculation, partial correspondence and missing results.

5.1 Quantitative comparison and sensitivity

The paper reports a style-ratio range of 0.4–42.4 for 63 events. The present range is 0.03879–463.9. The lower and upper endpoints differ by -0.3612 and 421.5, respectively. These are distribution-endpoint comparisons, not matched-event errors. Matching the number of events does not establish matching event membership or measurements.

The published distance transition near $D/w = 60$–100 and slope transition near 0.005 are comparison

targets only. They are not inputs to classification, masks, thresholds or geometry. The present independent geometry coverage cannot test either transition. The published individual-result and geometry tables are not imported.

Calculation	Minimum S_R	Maximum S_R
Primary spatial union	0.03879	463.9
Sum of image-part counts	0.03879	464
Alternative B14 onset	0.03879	463.9
Parent in avulsion years	0	30.86
Parent in all selected years	0	24.85

The temporal alternatives retain the same annual thresholds and avulsion windows; only the parent accumulation changes. They bracket specific interpretations of the written instruction to use the same time ranges where possible. They are not selected according to agreement with the paper. Every signature-specific result remains available.

The largest primary ratio illustrates the denominator sensitivity. For B10, the two calculated alternatives are B: 124780 avulsion pixels, 269 parent pixels and a ratio of 463.9; W: 520388 avulsion pixels, 25034 parent pixels and a ratio of 20.79. Minimizing avulsion area can choose a much larger ratio when that filter also detects far fewer parent-channel pixels. Both calculations use the same supplied mask geometry. This is an observed numerical consequence of the stated rule, not evidence that the larger ratio correctly measures floodplain disturbance.

The primary range fails to match the published range. Overlap accounting produces much smaller changes than parent-window selection. The experiment therefore establishes reproducibility of the stated downstream computation but does not establish reproduction of the paper's quantitative result. The missing independent geometry additionally prevents testing the proposed downstream transition.

For a limited matched-event check, Figure 5 explicitly prints $S_R = 0.5$ for V15 and 14.9 for V11. These two numerical annotations were read only after the calculations, without extracting plotted coordinates or image areas. They enter this comparison section only; the numerical analysis never reads them.

For V15, the recomputed ratio is 2.77, a difference of 2.27 (454.0 percent) relative to the printed value.

For V11, the recomputed ratio is 28.66, a difference of 13.76 (92.3 percent) relative to the printed value.

Both calculations rank V11 above V15, but the magnitudes disagree. The brightness-selected V15 fingerprint is visibly sparse compared with the published depiction. The B12 full-event mosaic covers a larger domain than the published central-part view. Its independently chosen detail and annual views do not recover the original validation window; they demonstrate the calculation only. Display extents were refined after comparison, without changing any numerical input or geometry observation. The published images are absent from the generated figures and repository.

A Figure-by-figure record

Panel	Status	Evidence and limit
1a	reproduced	Permitted coordinates and independently acquired terrain/vector geography. Geographic content reproduced; cartographic design differs.
1b	reproduced	Permitted coordinates and independently acquired terrain/vector geography. Geographic content reproduced; cartographic design differs.

Panel	Status	Evidence and limit
1c	reproduced	Permitted coordinates and independently acquired terrain/vector geography. Geographic content reproduced; cartographic design differs.
1d	reproduced	Permitted coordinates and independently acquired terrain/vector geography. Geographic content reproduced; cartographic design differs.
1 locator	reproduced	Permitted coordinates and independently acquired terrain/vector geography. Geographic content reproduced; cartographic design differs.
2a	partial	Independent cumulative raster from processed images and supplied masks. Timing, mask/signature convention and spatial correspondence limit equality claims.
2b	partial	Independent cumulative raster from processed images and supplied masks. Timing, mask/signature convention and spatial correspondence limit equality claims.
2c upper 2007	not reproduced	Archive filename inventory checked. Separate WorldView validation scenes are absent.
2c upper 2010	not reproduced	Archive filename inventory checked. Separate WorldView validation scenes are absent.
2c upper 2013	not reproduced	Archive filename inventory checked. Separate WorldView validation scenes are absent.
2c upper 2017	not reproduced	Archive filename inventory checked. Separate WorldView validation scenes are absent.
2c lower 2007	partial	Annual detections recomputed from the selected native band. Independent display extent; exact reference-panel pixel agreement not measured.
2c lower 2010	partial	Annual detections recomputed from the selected native band. Independent display extent; exact reference-panel pixel agreement not measured.
2c lower 2013	partial	Annual detections recomputed from the selected native band. Independent display extent; exact reference-panel pixel agreement not measured.
2c lower 2017	partial	Annual detections recomputed from the selected native band. Independent display extent; exact reference-panel pixel agreement not measured.
3a	not reproduced	Width observations attempted for four examples. No independently traced mountain-front distances; no distance regression.
3b	partial	Independent elevation profiles for illustrative examples. Insufficient coverage to reproduce the 63-event slope relationship.
4	partial	Four independent visual classifications joined to computed ratios. Small prespecified subset; no population-wide three-class distribution.
5a	partial	Independent geographic locator for V11 and V15. Intervening river course and physiographic mountain-front boundary not traced.
5b	partial	Supplied TCT image rendered with a documented false-color stretch. Corresponding 2019 Google Earth imagery is absent; different observation date.
5c	partial	Independent cumulative raster from processed images and supplied masks. Timing, mask/signature convention and spatial correspondence limit equality claims.
5d	partial	Supplied TCT image rendered with a documented false-color stretch. Corresponding 2019 Google Earth imagery is absent; different observation date.
5e	partial	Independent cumulative raster from processed images and supplied masks. Timing, mask/signature convention and spatial correspondence limit equality claims.
S1a	partial	Requested-year processed components independently rendered. TCT false color cannot reconstruct the original conventional-color imagery.
S1b	partial	Requested-year processed components independently rendered. TCT false color cannot reconstruct the original conventional-color imagery.
S1c	partial	Requested-year processed components independently rendered. TCT false color cannot reconstruct the original conventional-color imagery.

Panel	Status	Evidence and limit
S1d	partial	Requested-year processed components independently rendered. TCT false color cannot reconstruct the original conventional-color imagery.
S2a	partial	Independent cumulative raster from processed images and supplied masks. Timing, mask/signature convention and spatial correspondence limit equality claims.
S2b	partial	Independent cumulative raster from processed images and supplied masks. Timing, mask/signature convention and spatial correspondence limit equality claims.
S2c	not reproduced	Archive filename inventory checked. Separate high-resolution validation scene absent.
S3a	partial	Independent example slopes and visual morphology. No complete morphology-stratified population.
S3b	not reproduced	Geometric data requirements explicitly assessed. No mountain-front distances.

B Computed event table

Event	Filter	Avulsion pixels	Parent pixels	S_R
B1	B	988498	50717	19.49
B10	B	124780	269	463.9
B11	B	649	347	1.87
B12	B	106667	2358	45.24
B13	B	297	346	0.8584
B14	B	392743	8212	47.83
B15	B	201	1143	0.1759
B2	B	156	290	0.5379
B3	B	82	451	0.1818
B4	B	198	13	15.23
B6	B	1036	891	1.163
B7	B	101	162	0.6235
B8	W	229	137	1.672
B9	B	33265	2964	11.22
G1	B	8688	5847	1.486
G10	B	22	63	0.3492
G11	W	260	193	1.347
G12	B	169	19	8.895
G13	B	669	18	37.17
G14	B	140	297	0.4714
G15	B	127	92	1.38
G16	W	5	10	0.5
G17	B	35	32	1.094
G18	B	57	97	0.5876
G19	B	375	180	2.083
G2	B	1124	11	102.2
G20	B	501	555	0.9027
G21	B	411	563	0.73
G22	B	377	592	0.6368
G23	B	9	232	0.03879
G24	B	391	480	0.8146
G25_1	B	95	12	7.917
G25_2	B	87	118	0.7373

Event	Filter	Avulsion pixels	Parent pixels	S_R
G26	W	63	77	0.8182
G3	B	99	159	0.6226
G4	B	591	313	1.888
G5	B	1171	171	6.848
G6	B	365	174	2.098
G7	B	96	411	0.2336
G8	B	3190	38	83.95
G9	B	132	119	1.109
H1	B	21764	37876	0.5746
H2	B	3634	3533	1.029
H3	W	50	63	0.7937
H4	W	115	188	0.6117
H5	B	33145	36400	0.9106
H6	B	12000	1942	6.179
V1	B	270	170	1.588
V10	B	44	10	4.4
V11	B	14558	508	28.66
V12_1	B	87	114	0.7632
V12_2	B	376	7	53.71
V12_3	B	22	306	0.0719
V13	B	59	1101	0.05359
V14	B	43914	155	283.3
V15	B	482	174	2.77
V4	B	95	30	3.167
V5_1	B	138	295	0.4678
V5_2	B	22	2	11
V6	B	1931	2395	0.8063
V7	B	2795	139	20.11
V8	B	579	573	1.01
V9	B	2227	2944	0.7565

C Independent illustrative geometry

Event	Width (m)	Class	Reach (km)	OLS slope	Endpoint slope
B11	unmeasured	3	0	undefined	undefined
B12	128.3	3	4.69	-0.0004218	0.0004274
V11	unmeasured	3	7.686	0.001048	0.0007813
V15	unmeasured	1	7.983	0.007584	0.007393

Classes are independent coarse observations: 1, multiple visible threads at all sections; 2, mixed single/multiple sections; 3, one visible thread at all sections. Width estimates require all three sections; missing values are retained. OLS and endpoint columns include unsuccessful slope attempts.

References

[1] Valenza, J. M., Edmonds, D. A., Hwang, T. and Roy, S. (2020). Downstream changes in river avulsion style are related to channel morphology. *Nature Communications* 11, 2116. https://doi.org/10.1038/s41467-020-15859-9.

[2] Valenza, J., Edmonds, D., Hwang, T. and Roy, S. (2020). Downstream changes in river avulsion style are related to channel morphology: Data and Scripts. Indiana University DataCORE. https://hdl.handle.net/2022/25284; https://datacore.iu.edu/concern/data_sets/4x51hk339. Only the processed imagery, masks and permitted interpretation metadata are inputs to this experiment.

[3] Terrain Tiles public elevation archive. Provider attribution and source metadata are recorded in `data/provenance/terrain.json`. https://registry.opendata.aws/terrain-tiles/.

[4] Natural Earth, 1:110 million land polygons. https://www.naturalearthdata.com/downloads/110m-physical-vectors/110m-land/. Downloaded file identity is recorded in `data/provenance/land.json`.

[5] First independent reproduction, source and compiled report: https://github.com/RianMontagh/valenza-2020-independent-reproduction.